MSG No. 194.267 $P6'_3/mmc'$ [Type III, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3'm.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 5: Wyckoff site: `4e`, site symmetry: `3m.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12]`
2	$\left[0, 0, \frac{1}{2} - z\right]$	`[4,5,6,7,8,9]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$[0, 0, -z]$	`[16,17,18,19,20,21]`

Table 6: Wyckoff site: `4f`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[16,17,18,19,20,21]`

Table 7: Wyckoff site: 6g, site symmetry: .2'/m.

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,10,16,19]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,11,17,20]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,12,18,21]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[4,7,13,22]
5	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,8,14,23]
6	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[6,9,15,24]

Table 8: Wyckoff site: 6h, site symmetry: mm2

No.	position	mapping
1	$\left[x, 2x, \frac{1}{4}\right]$	[1,5,8,10]
2	$\left[-2x, -x, \frac{1}{4}\right]$	[2,6,9,11]
3	$\left[x, -x, \frac{1}{4}\right]$	[3,4,7,12]
4	$\left[-x, x, \frac{3}{4}\right]$	[13,18,21,22]
5	$\left[-x, -2x, \frac{3}{4}\right]$	[14,16,19,23]
6	$\left[2x, x, \frac{3}{4}\right]$	[15,17,20,24]

Table 9: Wyckoff site: 12i, site symmetry: .2'.

No.	position	mapping
1	$[x, 0, 0]$	[1,16]
2	$[0, x, 0]$	[2,17]
3	$[-x, -x, 0]$	[3,18]
4	$\left[x, x, \frac{1}{2}\right]$	[4,13]
5	$\left[-x, 0, \frac{1}{2}\right]$	[5,14]
6	$\left[0, -x, \frac{1}{2}\right]$	[6,15]
7	$\left[-x, -x, \frac{1}{2}\right]$	[7,22]
8	$\left[x, 0, \frac{1}{2}\right]$	[8,23]
9	$\left[0, x, \frac{1}{2}\right]$	[9,24]
10	$[-x, 0, 0]$	[10,19]
11	$[0, -x, 0]$	[11,20]
12	$[x, x, 0]$	[12,21]

Table 10: Wyckoff site: 12j, site symmetry: m..

No.	position	mapping
1	$\left[x, y, \frac{1}{4}\right]$	[1,8]
2	$\left[-y, x-y, \frac{1}{4}\right]$	[2,9]
3	$\left[-x+y, -x, \frac{1}{4}\right]$	[3,7]
4	$\left[x, x-y, \frac{1}{4}\right]$	[4,12]
5	$\left[-x+y, y, \frac{1}{4}\right]$	[5,10]
6	$\left[-y, -x, \frac{1}{4}\right]$	[6,11]
7	$\left[x-y, x, \frac{3}{4}\right]$	[13,21]

continued ...

Table 10

No.	position	mapping
8	$\left[-x,\ -y,\ \frac{3}{4}\right]$	[14,19]
9	$\left[y,\ -x+y,\ \frac{3}{4}\right]$	[15,20]
10	$\left[x-y,\ -y,\ \frac{3}{4}\right]$	[16,23]
11	$\left[y,\ x,\ \frac{3}{4}\right]$	[17,24]
12	$\left[-x,\ -x+y,\ \frac{3}{4}\right]$	[18,22]

Table 11: Wyckoff site: 12k, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,10]
2	$[-2x,\ -x,\ z]$	[2,11]
3	$[x,\ -x,\ z]$	[3,12]
4	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	[4,7]
5	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	[5,8]
6	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	[6,9]
7	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	[13,22]
8	$\left[-x,\ -2x,\ z+\frac{1}{2}\right]$	[14,23]
9	$\left[2x,\ x,\ z+\frac{1}{2}\right]$	[15,24]
10	$[-x,\ -2x,\ -z]$	[16,19]
11	$[2x,\ x,\ -z]$	[17,20]
12	$[-x,\ x,\ -z]$	[18,21]

Table 12: Wyckoff site: 24l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[6]
7	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[8]
9	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[9]
10	$[-x+y,\ y,\ z]$	[10]
11	$[-y,\ -x,\ z]$	[11]
12	$[x,\ x-y,\ z]$	[12]
13	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	[13]
14	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	[15]
16	$[x-y,\ -y,\ -z]$	[16]
17	$[y,\ x,\ -z]$	[17]
18	$[-x,\ -x+y,\ -z]$	[18]
19	$[-x,\ -y,\ -z]$	[19]

continued ...

Table 12

No.	position	mapping
20	$[y,\ -x+y,\ -z]$	[20]
21	$[x-y,\ x,\ -z]$	[21]
22	$\left[-x,\ -x+y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\ -y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[24]